

Control variates cancel the leading client-heterogeneity drift in a federated two-step round

Companion proof for: [pleyva2004/communication-efficient-learning-of-deep-networks](#)

Improvement slug: `heterogeneity-control-variate`

Abstract

The math deep dive (eq. (3.5)) showed that running two local gradient-descent steps per client and then averaging deviates from two genuine centralized gradient-descent steps by a leading term $\eta^2 \text{Cov}_k(H_k, g_k) + O(\eta^3)$, the client-weighted covariance between local Hessians H_k and local gradients g_k . This is the “client-drift” object that later federated-learning theory (e.g. Karimireddy et al. 2020, SCAFFOLD) is explicitly designed to remove. Here we prove, by carrying the *same* Taylor expansion but replacing each local gradient $g_{\text{local}}(w)$ with the control-variate-corrected direction $g_{\text{local}}(w) - c_k + c$ (where c_k is a fixed estimate of client k ’s gradient at the shared point w_t and $c = \sum_k \alpha_k c_k$ is its client-weighted mean), that the entire order- η^2 covariance term cancels. Consequently the corrected averaged two-step round matches two centralized gradient-descent steps up to $O(\eta^3)$, recovering, to leading order, the per-round progress of centralized full-batch gradient descent even on an arbitrarily heterogeneous (non-IID) partition.

Setup and notation

We adopt verbatim the setup of the math deep dive. The global objective is the finite sum $f(w) = \sum_{k=1}^K \alpha_k F_k(w)$ with mixture weights $\alpha_k := n_k/n \geq 0$, $\sum_{k=1}^K \alpha_k = 1$, where $F_k(w) = \frac{1}{n_k} \sum_{i \in P_k} f_i(w)$ is client k ’s local objective over its example set P_k of size n_k ; this is the exact algebraic identity (1.1) of the deep dive, valid for any partition. All clients begin a round at the common shared point $w_t \in \mathbb{R}^d$ (the load-bearing shared-initialization assumption). We fix the round and write everything relative to w_t .

We work in the *quadratic / second-order local model* used to derive eq. (3.5): each F_k is twice continuously differentiable in a neighborhood of w_t , and we expand each local gradient to first order about w_t with a uniform $O(\cdot)$ remainder. Concretely, writing $g_k := \nabla F_k(w_t)$ and $H_k := \nabla^2 F_k(w_t)$, for any displacement δ of size $O(\eta)$ Taylor’s theorem with remainder gives

$$\nabla F_k(w_t + \delta) = g_k + H_k \delta + R_k(\delta), \quad \|R_k(\delta)\| = O(\|\delta\|^2), \quad (1)$$

where the constant hidden in $O(\|\delta\|^2)$ is uniform over the finitely many clients $k \in \{1, \dots, K\}$ (take the maximum over k of the per-client constants). Because every displacement we encounter is $O(\eta)$, every such remainder contributes at order η^2 *inside* a term that is already multiplied by η , i.e. at order η^3 in the position iterate; this is exactly the bookkeeping that makes the deep dive’s $O(\eta^3)$ statement precise. The learning rate is $\eta > 0$, taken small enough that the round’s displacements stay in the neighborhood where (1) holds.

Definition 1 (Working notation). Throughout the round:

- $g_k := \nabla F_k(w_t) \in \mathbb{R}^d$ — client k ’s local gradient at the shared point w_t .

- $H_k := \nabla^2 F_k(w_t) \in \mathbb{R}^{d \times d}$ — client k 's local Hessian at w_t .
- $\nabla f := \nabla f(w_t) = \sum_{k=1}^K \alpha_k g_k$ — the global gradient at w_t ; the equality is the exact gradient-aggregation identity (2.2) of the deep dive, obtained by differentiating $f = \sum_k \alpha_k F_k$.
- $H := \nabla^2 f(w_t) = \sum_{k=1}^K \alpha_k H_k$ — the global Hessian at w_t , again by differentiating the mixture identity twice.
- $\text{Cov}_k(H_k, g_k) := \sum_{k=1}^K \alpha_k H_k g_k - \left(\sum_{k=1}^K \alpha_k H_k \right) \left(\sum_{k=1}^K \alpha_k g_k \right) = \sum_{k=1}^K \alpha_k H_k g_k - H \nabla f$ — the client-weighted covariance between local Hessians and local gradients (a vector in $\mathbb{R}^d$), with the α_k acting as the client distribution. This is the leading deviation term of eq. (3.5).
- **Control variates.** Each client k carries a fixed vector c_k that estimates its own local gradient direction at the shared point w_t ; in the static, exact-estimate regime analyzed here we take $c_k := g_k = \nabla F_k(w_t)$, held *constant* for the whole round (it is computed once at w_t and not refreshed between the two local steps). The server control variate is the client-weighted mean $c := \sum_{k=1}^K \alpha_k c_k = \sum_{k=1}^K \alpha_k g_k = \nabla f$, also held constant for the round.
- **Corrected local direction.** At any iterate w , instead of stepping along the raw local gradient $g_{\text{local}}(w) := \nabla F_k(w)$, client k steps along the control-variate-corrected direction

$$\tilde{g}_k(w) := \nabla F_k(w) - c_k + c. \quad (2)$$

Theorem

Theorem 2 (Control variates cancel the order- η^2 heterogeneity drift). *Fix a round at the shared point w_t and adopt the quadratic / second-order local model (1) with static, exact control variates $c_k = g_k$ and $c = \sum_k \alpha_k g_k = \nabla f$. Let each client run two local steps using the corrected direction (2),*

$$w_{(0)}^k = w_t, \quad w_{(1)}^k = w_{(0)}^k - \eta \tilde{g}_k(w_{(0)}^k), \quad w_{(2)}^k = w_{(1)}^k - \eta \tilde{g}_k(w_{(1)}^k),$$

and let the server form the client-weighted average $\bar{w}_{t+2} := \sum_{k=1}^K \alpha_k w_{(2)}^k$. Let $G^{(2)}(w_t)$ denote two genuine centralized gradient-descent steps on f from w_t , namely $u_1 = w_t - \eta \nabla f$, $G^{(2)}(w_t) = u_1 - \eta \nabla f(u_1)$. Then the order- η^2 deviation $\eta^2 \text{Cov}_k(H_k, g_k)$ of eq. (3.5) vanishes to first order: the corrected averaged round equals two centralized steps up to $O(\eta^3)$,

$$\bar{w}_{t+2} = w_t - 2\eta \nabla f + \eta^2 H \nabla f + O(\eta^3) = G^{(2)}(w_t) + O(\eta^3),$$

whereas the uncorrected FedAvg round (using g_{local} directly) satisfies $\bar{w}_{t+2}^{\text{FedAvg}} = G^{(2)}(w_t) + \eta^2 \text{Cov}_k(H_k, g_k) + O(\eta^3)$. In particular the leading discrepancy is reduced from $\Theta(\eta^2)$ (generically nonzero under client heterogeneity) to $O(\eta^3)$.

Proof

Proof. The proof carries the same Taylor expansion as the deep dive's derivation of eq. (3.5), but applied to the corrected direction (2). We proceed in five steps: (i) compute the corrected first step; (ii) Taylor-expand the corrected second step; (iii) assemble the corrected two-step endpoint per client; (iv) average over clients; (v) compare against two centralized steps and against the uncorrected FedAvg round, exhibiting the cancellation of the covariance term.

Step (i): the corrected first step is identical across clients and equals one centralized step. Evaluate (2) at the shared start $w_{(0)}^k = w_t$. By definition $\nabla F_k(w_t) = g_k$ and, in the static exact-estimate regime, $c_k = g_k$, so the raw local gradient and the client's own control variate cancel exactly:

$$\tilde{g}_k(w_t) = \underbrace{\nabla F_k(w_t)}_{=g_k} - \underbrace{c_k}_{=g_k} + c = g_k - g_k + c = c = \nabla f. \quad (3)$$

This is the crux of the correction: the term $-c_k$ removes the client's own gradient and $+c$ substitutes the *global* direction ∇f , so the first corrected step is the same for every client and equals exactly one centralized gradient-descent step,

$$w_{(1)}^k = w_t - \eta \tilde{g}_k(w_t) = w_t - \eta \nabla f \quad \text{for every } k. \quad (4)$$

In particular $w_{(1)}^k$ does not depend on k ; we may write it as $u_1 := w_t - \eta \nabla f$, which is precisely the first centralized iterate.

Step (ii): Taylor-expand the corrected second direction. We now evaluate (2) at $w_{(1)}^k = w_t - \eta \nabla f$. The displacement from w_t is $\delta := -\eta \nabla f$, which is $O(\eta)$, so (1) applies with remainder $\|R_k(\delta)\| = O(\eta^2)$:

$$\nabla F_k(w_t - \eta \nabla f) = g_k + H_k(-\eta \nabla f) + R_k(-\eta \nabla f) = g_k - \eta H_k \nabla f + O(\eta^2). \quad (5)$$

Subtract the static control variate $c_k = g_k$ and add $c = \nabla f$. The constant local gradient g_k cancels against $-c_k = -g_k$ a *second* time — this is the structural reason the correction keeps working past the first step — leaving

$$\tilde{g}_k(w_{(1)}^k) = (g_k - \eta H_k \nabla f + O(\eta^2)) - g_k + \nabla f = \nabla f - \eta H_k \nabla f + O(\eta^2). \quad (6)$$

Notice that the leading $O(1)$ part of the corrected second direction is again the global ∇f , with the *only* client-dependent piece appearing at order η through the client's own Hessian H_k acting on the *global* gradient ∇f — not on the local g_k . This single change (the matrix H_k now multiplies ∇f rather than g_k) is what will make the covariance collapse in Step (v).

Step (iii): assemble the corrected two-step endpoint per client. Combine (4) and (6):

$$\begin{aligned} w_{(2)}^k &= w_{(1)}^k - \eta \tilde{g}_k(w_{(1)}^k) \\ &= (w_t - \eta \nabla f) - \eta(\nabla f - \eta H_k \nabla f + O(\eta^2)) \\ &= w_t - \eta \nabla f - \eta \nabla f + \eta^2 H_k \nabla f + \eta \cdot O(\eta^2) \\ &= w_t - 2\eta \nabla f + \eta^2 H_k \nabla f + O(\eta^3). \end{aligned} \quad (7)$$

Here the term $\eta \cdot O(\eta^2) = O(\eta^3)$ absorbs the propagated Taylor remainder: the $O(\eta^2)$ error in the second direction (6) is multiplied by the step size η , hence enters the position iterate (7) only at order η^3 , exactly as claimed. This is the corrected analogue of the deep dive's uncorrected endpoint $w_{(2),\text{FedAvg}}^k = w_t - 2\eta g_k + \eta^2 H_k g_k + O(\eta^3)$: the $O(1)$ term has moved from the client-specific $-2\eta g_k$ to the global $-2\eta \nabla f$, and the $O(\eta^2)$ term from $H_k g_k$ to $H_k \nabla f$.

Step (iv): average over clients. Apply the server's client-weighted average $\sum_k \alpha_k(\cdot)$ to (7).

Linearity of the sum and $\sum_k \alpha_k = 1$ give, term by term,

$$\begin{aligned}\bar{w}_{t+2} &= \sum_{k=1}^K \alpha_k \left(w_t - 2\eta \nabla f + \eta^2 H_k \nabla f + O(\eta^3) \right) \\ &= \left(\underbrace{\sum_k \alpha_k}_{=1} \right) w_t - 2\eta \left(\underbrace{\sum_k \alpha_k}_{=1} \right) \nabla f + \eta^2 \left(\underbrace{\sum_k \alpha_k H_k}_{=H} \right) \nabla f + O(\eta^3) \\ &= w_t - 2\eta \nabla f + \eta^2 H \nabla f + O(\eta^3).\end{aligned}\tag{8}$$

Two facts make the averaging clean. First, w_t and ∇f are constants with respect to the index k (the global gradient $\nabla f = \sum_j \alpha_j g_j$ does not depend on the outer k), so they pass through the average unchanged via $\sum_k \alpha_k = 1$. Second, the only k -dependent object surviving to order η^2 is the Hessian H_k , and it is multiplied by the k -independent vector ∇f ; hence the average of $H_k \nabla f$ is exactly $(\sum_k \alpha_k H_k) \nabla f = H \nabla f$, with no cross term. The $O(\eta^3)$ remainders, being a finite α_k -weighted ($\sum_k \alpha_k = 1$, $\alpha_k \geq 0$) average of $O(\eta^3)$ vectors with uniform constant, remain $O(\eta^3)$.

Step (v): compare to two centralized steps, and exhibit the cancellation. Two genuine centralized gradient-descent steps on f from w_t give, by the same Taylor expansion (1) applied to ∇f with global Hessian $H = \nabla^2 f(w_t)$,

$$\begin{aligned}u_1 &= w_t - \eta \nabla f, \\ G^{(2)}(w_t) &= u_1 - \eta \nabla f(u_1) = u_1 - \eta(\nabla f - \eta H \nabla f + O(\eta^2)) \\ &= w_t - 2\eta \nabla f + \eta^2 H \nabla f + O(\eta^3),\end{aligned}\tag{9}$$

which is identical, term by term up to $O(\eta^3)$, to the corrected averaged round (8). Subtracting,

$$\bar{w}_{t+2} - G^{(2)}(w_t) = O(\eta^3).\tag{10}$$

To see precisely what was cancelled, repeat Step (iv) for the *uncorrected* FedAvg endpoint $w_{(2),\text{FedAvg}}^k = w_t - 2\eta g_k + \eta^2 H_k g_k + O(\eta^3)$ of the deep dive. Averaging,

$$\bar{w}_{t+2}^{\text{FedAvg}} = w_t - 2\eta \sum_k \alpha_k g_k + \eta^2 \sum_k \alpha_k H_k g_k + O(\eta^3) = w_t - 2\eta \nabla f + \eta^2 \sum_k \alpha_k H_k g_k + O(\eta^3),\tag{11}$$

using $\sum_k \alpha_k g_k = \nabla f$. Subtracting the centralized expansion (9) from (11), the w_t and $-2\eta \nabla f$ terms cancel and the leading deviation is

$$\bar{w}_{t+2}^{\text{FedAvg}} - G^{(2)}(w_t) = \eta^2 \left(\sum_{k=1}^K \alpha_k H_k g_k - H \nabla f \right) + O(\eta^3) = \eta^2 \text{Cov}_k(H_k, g_k) + O(\eta^3),\tag{12}$$

which is exactly eq. (3.5) of the deep dive, by the definition of Cov_k recorded in the Working notation.

The cancellation is now transparent at the level of the order- η^2 coefficient. The uncorrected coefficient is $\sum_k \alpha_k H_k g_k$, in which the matrix H_k multiplies the *client-specific* gradient g_k ; its average does *not* factor, leaving the residual $\sum_k \alpha_k H_k g_k - H \nabla f = \text{Cov}_k(H_k, g_k)$, which is generically nonzero whenever the clients are heterogeneous (different H_k correlated with different g_k). The corrected coefficient, by contrast, is $\sum_k \alpha_k H_k \nabla f$, in which H_k multiplies the *global* gradient ∇f — a constant in k — so the average factors exactly as $(\sum_k \alpha_k H_k) \nabla f = H \nabla f$, leaving no residual. Equivalently, the difference between the corrected and uncorrected order- η^2 terms is

$$\eta^2 \sum_{k=1}^K \alpha_k H_k (\nabla f - g_k) = \eta^2 \left(H \nabla f - \sum_{k=1}^K \alpha_k H_k g_k \right) = -\eta^2 \text{Cov}_k(H_k, g_k),\tag{13}$$

i.e. the control variate replaces the local gradient g_k that H_k would have acted on by the global gradient ∇f inside the second step, and the resulting change is precisely $-\eta^2 \text{Cov}_k(H_k, g_k)$, the exact negative of the FedAvg drift (12). Adding this to (12) sends the leading deviation to zero, leaving only $O(\eta^3)$, which is (10).

This proves both displayed equalities of Theorem 2: the corrected averaged round matches two centralized steps to $O(\eta^3)$, while the uncorrected round retains the $\Theta(\eta^2)$ heterogeneity term. The order- η^2 deviation $\eta^2 \text{Cov}_k(H_k, g_k)$ therefore vanishes to first order under the control-variate correction. $\square$

Discussion

What this gives us. The leading-order client-heterogeneity penalty of running more than one local step — the term $\eta^2 \text{Cov}_k(H_k, g_k)$ that the deep dive identified as the entire $O(\eta^2)$ price of the second local step on non-IID data — is removed exactly by the control-variate correction $g_{\text{local}}(w) \mapsto g_{\text{local}}(w) - c_k + c$. After the fix, a corrected two-step federated round reproduces two centralized full-batch gradient-descent steps up to $O(\eta^3)$, i.e. the per-round progress is, to leading order, indistinguishable from centralized training *regardless of how heterogeneous the partition is*. The mechanism is explicit and local: $-c_k$ subtracts the client’s own gradient and $+c$ inserts the global one, so each corrected step is centered on ∇f rather than on the client’s biased g_k . This is the formal content behind importing SCAFFOLD-style control variates (Karimireddy et al. 2020) into FedAvg, and it is the leading-order counterpart of the proximal anchor used by FedProx (Li et al. 2020), which attacks the same drift object by a different route.

What this does not give us. The result is a *leading-order*, single-round, per-step statement, not a convergence theorem: it does not by itself bound the number of communication rounds, prove descent of f , or rule out divergence; it only equates one corrected two-step round with two centralized steps to $O(\eta^3)$. It says nothing about the $O(\eta^3)$ and higher terms, which still carry genuine client-dependence (third derivatives of F_k , the propagated remainder $\eta \cdot O(\eta^2)$ of Step (iii)); the correction zeroes the η^2 coefficient, not the whole gap. The analysis is deterministic and full-batch, with $g_{\text{local}} = \nabla F_k$ exact; it does not address stochastic minibatch noise (where c_k, c are themselves noisy estimates and the cancellation holds only in expectation), partial client participation, or the extra per-client state and the control-variate refresh that a multi-round deployment requires. The proof also assumes the standard FedAvg shared initialization $w_{(0)}^k = w_t$; without it even the first step (4) would differ across clients.

Where it would break. Two assumptions are load-bearing. (1) *Static, accurate control variates.* The cancellation in Steps (i)–(ii) relied twice on $c_k = g_k = \nabla F_k(w_t)$ holding for the whole round. If c_k is *stale* — estimated at an earlier global model $w_{t'} \neq w_t$, as happens with intermittent participation — then $c_k = \nabla F_k(w_{t'}) = g_k - H_k(w_t - w_{t'}) + \dots$, and the leftover $-c_k + c$ no longer cancels g_k cleanly: a residual proportional to the staleness gap $w_t - w_{t'}$ re-enters at order η in the corrected direction and hence at order η^2 in the iterate, partially restoring a drift term. The cleaner $c_k = g_k$ the estimate, the more of the η^2 covariance is removed; an arbitrarily stale c_k recovers no benefit. (2) *Mild nonlinearity (validity of the quadratic model).* The whole argument truncates the Taylor expansion (1) at first order in the gradient (second order in F_k). Under strong nonlinearity — large third derivatives $\nabla^3 F_k$, or a learning rate η large enough that the round’s displacements leave the neighborhood where (1) holds — the discarded $O(\eta^3)$ terms are no longer negligible and carry their own uncanceled client-heterogeneity contributions; the control variate is

then guaranteed to remove only the η^2 piece, and the practical benefit shrinks as η grows or the local landscapes become more curved/heterogeneous at third order. Pushing the number of local steps well beyond two compounds this: each additional step adds higher-order remainders that the single first-order control variate does not address.

Empirical companion. This proof is the validation file for the improvement `heterogeneity-control-variate` in `05-improvements.tex` (subsection M.1). If a measurement-mode prototype also exists at `improvements/heterogeneity-control-variate.py`, it serves as a sanity check, not as evidence — the theorem above is the load-bearing claim.